Seminar 10

3.12.2025

Let $f: V \to W$ homomorphism.

Let $B = (v_1, \dots, v_m)$ basis of V.

$B' = (w_1, \dots, w_m)$ basis of W.

$$[f]_B = \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ f(v_1) & f(v_2) & \vdots & f(v_m) \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$[f]_{BB'}$ = if the columns containing $f(v_i)$ are expressed as linear comb. of vectors from V'.

1.) $f \in End_{\mathbb{R}}(\mathbb{R}^3)$

$f(x,y,z) = (x+y, y-z, 2x+y+z)$

$[f]_E$, E canonical basis of $\mathbb{R}^3$.

$E = (e_1, e_2, e_3) = ((1,0,0), (0,1,0), (0,0,1))$.

$f(e_1) = f(1,0,0) = (1,0,2)$.

$f(e_2) = f(0,1,0) = (1,1,1)$

$f(e_3) = f(0,0,1) = (0,-1,1)$

$$[f]_E = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} = [f]_{EE}.$$

2) $f \in Hom_{\mathbb{R}}(\mathbb{R}^3, \mathbb{R}_2)$.

$f(x,y) = (y,-x)$.

$B = (v_1, v_2, v_3)$

$v_1 = (1,1,0)$

$v_2 = (0,1,1)$

$v_3 = (1,0,1)$

$B' = (v_1', v_2')$

$v_1' = (1,1)$ E' of $\mathbb{R}^2$.

$v_2' = (1,-2)$

$[f]_{BE'} = ?$

$[f]_{BB'} = ?$

$$[f]_{BE'} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

$$[f]_{BB'} = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

③. $f \in Hom_{\mathbb{R}}(\mathbb{R}^3, \mathbb{R}^4)$

$f(e_1) = (1,2,3,4)$

$f(e_2) = (4,3,2,1)$

$f(e_3) = (-2,1,4,1)$

(e_1, e_2, e_3) of $\mathbb{R}^3$.

a) $f(v) = ?$, $(\forall) v \in \mathbb{R}^3$.

b) $[f]_E = ?$

c) a basis, dimension of $Ker f$, $Im f$.

we can express any vector $v \in \mathbb{R}^3$ using canonical vect

a) $v = \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3 \mid f$

$$f(v) = f(\alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3) \overset{hom.}{=} f(\alpha_1 e_1) + f(\alpha_2 e_2) + f(\alpha_3 e_3) =$$

$$= \alpha_1 f(e_1) + \alpha_2 f(e_2) + \alpha_3 f(e_3) = \alpha_1 (1,2,3,4) + \alpha_2 (4,3,2,1) + \alpha_3 (-2,1,4,1)$$

$$\Rightarrow f(v) = (\alpha_1 + 4\alpha_2 - 2\alpha_3,\ 2\alpha_1 + 3\alpha_2 + \alpha_3,\ 4\alpha_1 + 2\alpha_2 + 4\alpha_3,\ 4\alpha_1 + \alpha_2 + \alpha_3)$$

$, (\forall) v \in \mathbb{R}^3$.

b) $$[f]_E = [f]_{EE'} = \begin{bmatrix} 1 & 4 & -2 \\ 2 & 3 & 1 \\ 3 & 2 & 4 \\ 4 & 1 & 1 \end{bmatrix}$$

E: can. base of $\mathbb{R}^3$; E': can. base of $\mathbb{R}^4$.

c) $$\boxed{[f]_E \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}} \iff \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in Ker f$$

$$\iff \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 2 & 4 & 0 \\ 4 & 1 & 1 & 0 \end{array}\right] \xrightarrow[\substack{R_3 \leftarrow R_3 - 3R_1 \\ R_4 \leftarrow R_4 - 4R_1}]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & -10 & 10 & 0 \\ 0 & -15 & 9 & 0 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 \cdot \frac{1}{3}]{R_3 \leftarrow R_3 - 2R_2}$$

$$= \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -5 & 3 & 0 \end{array}\right] \xrightarrow{R_4 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & -5 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - R_2}$$

$\det = 10 \neq 0$

$$= \left[\begin{array}{ccc|c} 1 & 4 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{cases} -2z = 0 & \Rightarrow z = 0 \\ -5y + 5z = 0 & \Rightarrow y = 0 \\ x + 4y - 2z = 0 & \Rightarrow x = 0 \end{cases}$$

$\Rightarrow Ker f = \langle (0,0,0) \rangle$ basis $\Rightarrow \dim Ker f = 0$.

$\boxed{\dim \operatorname{Im} f = \cancel{\dim}\ \operatorname{rank} [f]_E}$

$\dim \operatorname{Im} f = 3$ (det... $= 10 \neq 0$)

a basis for $\operatorname{Im} f = \langle (1,4,-2), (0,-5,5), (0,0,-2) \rangle$.

④ $f \in \operatorname{End}_{\mathbb{R}}(\mathbb{R}^4)$

$$[f]_E = \begin{bmatrix} 1 & 1 & -3 & 2 \\ -1 & 1 & 1 & 4 \\ 2 & 1 & -5 & 1 \\ 1 & 2 & -4 & 5 \end{bmatrix}$$

a) $v = (1,4,1,-1) \overset{?}{\in} \operatorname{Ker} f$

b) basis and dim $\operatorname{Ker} f$, $\operatorname{Im} f$?

c) define f.

a) $v = (1,4,1,-1) \in \operatorname{Ker} f \iff f(v) = 0'$.

$\iff f(1,4,1,-1) = (0,0,0,0)$

$$\begin{bmatrix} 1 & 1 & -3 & 2 \\ -1 & 1 & 1 & 4 \\ 2 & 1 & -5 & 1 \\ 1 & 2 & -4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\iff \begin{bmatrix} 1+4-3-2 \\ \cancel{-1}+\cancel{4}+\cancel{1}-\cancel{4} \\ 2+4-5-1 \\ 1+8-4-5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \iff \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ true.}$$

$\Rightarrow v \in \operatorname{Ker} f$.

b)
$$\left[\begin{array}{cccc|c} 1 & 1 & -3 & 2 & 0 \\ -1 & 1 & 1 & 4 & 0 \\ 2 & 1 & -5 & 1 & 0 \\ 1 & 2 & -4 & 5 & 0 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1} \left[\begin{array}{cccc|c} 1 & 1 & -3 & 2 & 0 \\ 0 & 2 & -2 & 6 & 0 \\ 2 & 1 & -5 & 1 & 0 \\ 0 & 1 & -1 & 3 & 0 \end{array}\right]$$

$1-2$

$-5+6$

$1-$

$$\xrightarrow[R_2 \leftrightarrow R_3]{R_2 \leftarrow R_2 \cdot \frac{1}{2}} \left[\begin{array}{cccc|c} 1 & 1 & -3 & 2 & 0 \\ 2 & 1 & -5 & 1 & 0 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 3 & 0 \end{array}\right] \xrightarrow{R_4 \leftarrow R_4 - R_3} \left[\begin{array}{ccccc} 1 & 1 & -3 & 2 & 0 \\ 2 & 1 & -5 & 1 & 0 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 1 & -3 & 2 & 0 \\ 0 & -1 & 1 & -3 & 0 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 + R_2} \left[\begin{array}{cccc|c} 1 & 1 & -3 & 2 & 0 \\ 0 & -1 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \begin{cases} x+y-3z+2t=0 \\ -y+z-3t=0 \end{cases} \Rightarrow \begin{cases} x+y = 3z-2t \\ -y = 3t-z \end{cases} \Rightarrow \begin{cases} x+y = 3z-2t \\ y = z-3t. \end{cases}$$

$x+z-3t = 3z-2t \Rightarrow x = 2z+t$

$\mathrm{Ker} f = \{(2z+t, z-3t, z, t) \mid z, t \in \mathbb{R}\}$

$= \langle (2,1,1,0), (1,-3,0,1) \rangle$ basis.

$\Rightarrow \dim \mathrm{Ker} f = 2.$

$\mathrm{Im} f = \langle (1,1,-3,2), (0,-1,1,3) \rangle$ (from the matrix) $\rightarrow$ basis

$\Rightarrow \dim \mathrm{Im} f = 2.$

$$\boxed{\dim \mathrm{Ker} f + \dim \mathrm{Im} f = \dim(\text{domain})}$$
$$2 + 2 = 4 \checkmark$$

scalars. (x, y, z, t)

c) $\cancel{f(e_1)}$, $\cancel{f(e_2)}$, $\cancel{f(e_3)}$, $\cancel{f(}$

$v = (v_1, v_2, v_3, v_4)$ / f

$f(v) = f(v_1, v_2, v_3, v_4)$

$f(v_1e_1 + v_2e_2 + v_3e_3 + v_4e_4) =$

$= v_1 f(e_1) + v_2 f(e_2) + v_3 f(e_3) + v_4 f(e_4)$

$= v_1(1,-1,2,1) + v_2(1,1,1,2) + v_3(-3,1,-5,-4) +$

$+ v_4(2,4,1,5)$

$\Rightarrow (v_1 + v_2 - 3v_3 + 2v_4, -v_1 + v_2 + v_3 + 4v_4, 2v_1 + v_2 - 5v_3 + v_4,$

$v_1 + 2v_2 - 4v_3 + 5v_4)$

⑥. $f, g \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^2)$

$B = (v_1, v_2) = ((1,2), (1,3))$.

$B' = (v_1', v_2') = ((1,0), (2,1))$.

Exam

$[f]_B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}$

$[g]_{B'} = \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix}$

$[2f]_B, [f+g]_B, [f \circ g]_{B'} = ?$

$$\boxed{[\alpha\cdot f]_B = \alpha\cdot[f]_B}$$

$$[2f]_B = 2[f]_B = \begin{bmatrix} 2 & 4 \\ -2 & -2 \end{bmatrix} \checkmark$$

$$\boxed{[f+g]_B = [f]_B \pm [g]_B}$$

$g(x,y) = (\alpha x + \beta y\ ,\ ax + by)$

$g(v_1') = (-7,5) \Leftrightarrow g(1,0) = (-7,5) \Leftrightarrow (\alpha, a) = (-7,5).$

$g(v_2') = (-13,7) \Leftrightarrow g(2,1) = (-13,7) \Leftrightarrow (2\alpha+\beta,\ 2a+b) = (-13,7)$

$\Leftrightarrow (-14+\beta,\ 10+b) = (-13,7) \Rightarrow \beta = 1$

$b = -3$

$g(x,y) = (-7x+y,\ 5x-3y)$

$g(v_1) = g(1,2) = (-5,-1)$

$g(v_2) = g(1,3) = (-4,-4)$ $\Rightarrow [g]_B = \begin{bmatrix} -5 & -4 \\ -1 & -4 \end{bmatrix}.$

$$[f+g]_B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} + \begin{bmatrix} -5 & -4 \\ -1 & -4 \end{bmatrix} = \begin{bmatrix} -4 & -2 \\ -2 & -5 \end{bmatrix}. \checkmark$$

$$\boxed{[f\circ g]_{B'} = [f]_{B'}\cdot[g]_{B'}.}$$

$-1-(-1)$

$f(x,y) = (\alpha x + \beta y,\ ax+by) = (-x+y,\ -x)$

~~$f(v_1') = f($~~ $\begin{cases} f(v_1) = (1,-1) \\ f(v_2) = (2,-1) \end{cases} \Leftrightarrow \begin{cases} f(1,2) = (1,-1) \\ f(1,3) = (2,-1) \end{cases} \Rightarrow$

$\Rightarrow \begin{cases} \cancel{\alpha+2\beta=1}\ (\alpha+2\beta,\ a+2b) = (1,-1) \\ (\alpha+3\beta,\ a+3b) = (2,-1) \end{cases} \Rightarrow \begin{cases} \alpha+2\beta = 1 & (1) \\ a+2b = -1 & (2) \\ \alpha+3\beta = 2 & (3) \\ a+3b = -1. & (4) \end{cases}$

"$(1)-(3)$" $\Rightarrow -\beta = -1 \Rightarrow \beta = 1;\ \alpha = 1-2 = -1$

"$(2)-(4)$" $\Rightarrow -b = 0 \Rightarrow b = 0 \Rightarrow a = -1.$

$\begin{cases} \alpha = -1 \\ \beta = 1 \\ a = -1 \\ b = 0 \end{cases}$

$f(v_1) = f(1,2) = (1,-1) \Rightarrow (x+2\beta,\ a+2b) = (1,-1) \Rightarrow \begin{cases} x+2\beta = 1 \\ a+2b = -1 \end{cases}$

$f(v_2) = f(1,3) = (2,-1) \Rightarrow (x+3\beta,\ a+3b) = (2,-1) \Rightarrow \begin{cases} x+3\beta = 2 \\ a+3b = -1. \end{cases}$

$f(v_1') = f(1,0) = (-1,-1)$

$\Rightarrow [f]_{B'} = \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix}$

$f(v_2') = f(2,1) = (-1,-2)$

$$[f \circ g]_{B'} = [f]_{B'} \cdot [g]_{B'}$$

$$= \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ -3 & -1 \end{bmatrix}$$

(7) $f : \mathbb{R}^2 \to \mathbb{R}^2$.

$f(x,y) = (x\cos\alpha - y\sin\alpha,\ x\sin\alpha + y\cos\alpha)$.

$\alpha \in \mathbb{R}$.

$[f]_E = ?$ Prove that f automorphism over $\mathbb{R}^2$. ($f \in Aut_{\mathbb{R}}(\mathbb{R}^2)$)

$f(e_1) = f(1,0) = (\cos\alpha,\ \sin\alpha)$

$f(e_2) = f(0,1) = (-\sin\alpha,\ \cos\alpha)$

$$[f]_E = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$\boxed{f \in Aut_{\mathbb{R}}(\mathbb{R}^2) \overset{(m)}{\Longleftrightarrow} \det [f]_E \neq 0}$$

$$\begin{vmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{vmatrix} = \cos^2\alpha + \sin^2\alpha = 1 \neq 0 \Rightarrow f \in Aut_{\mathbb{R}}(\mathbb{R}^2)$$